

Netflix Dataset

Exploratory Data Analysis Report

Thiranex Internship — Week 3

Dataset: 8,775 titles | 6,112 Movies | 2,663 TV Shows

Netflix Dataset — Business KPI Overview

8775

Total Titles

6112

Movies

2663

TV Shows

86

Countries

36

Unique Genres

73

Years Covered

2019

Peak Year

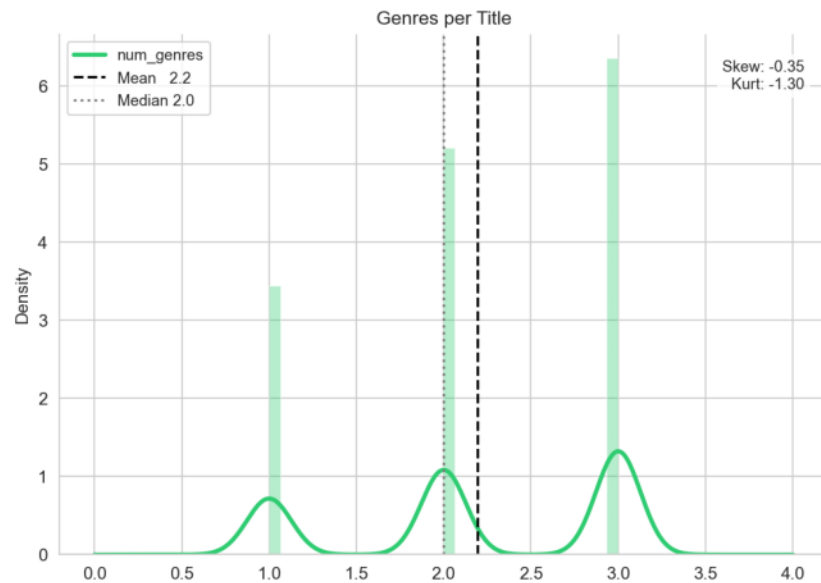
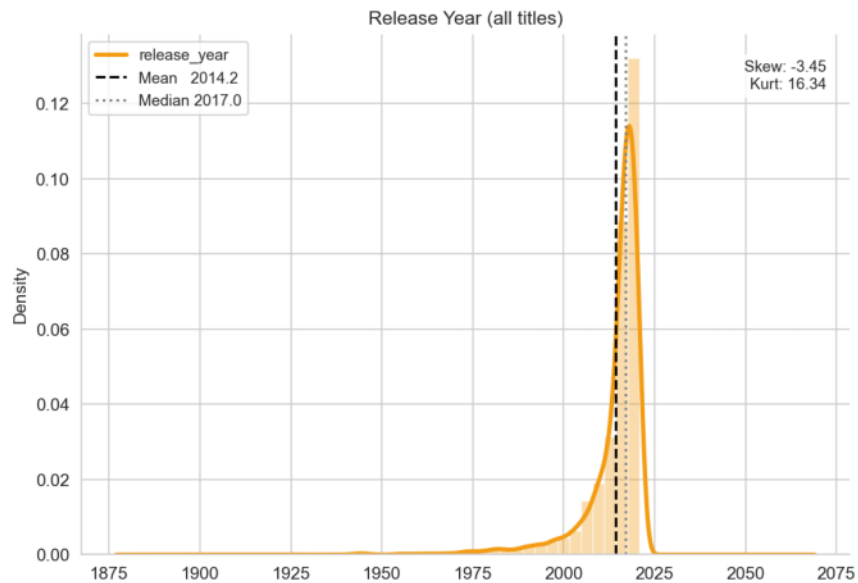
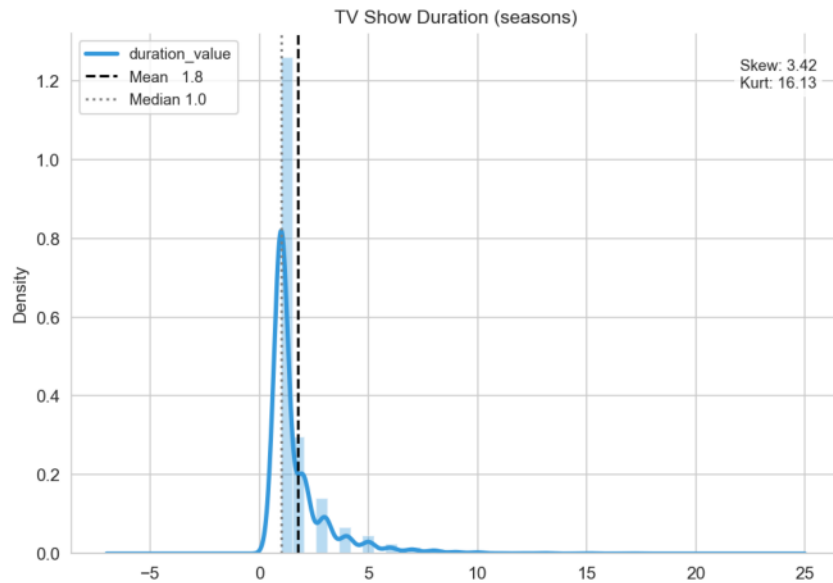
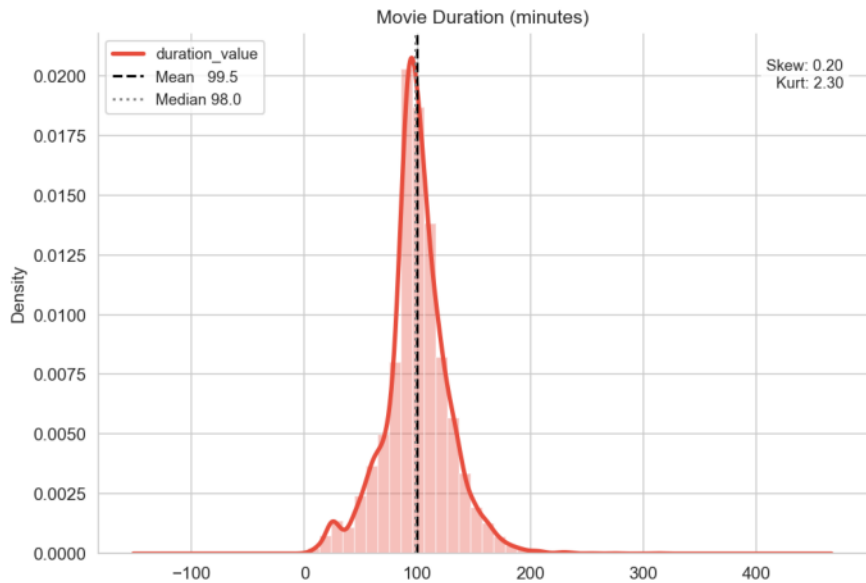
100

Avg Movie (min)

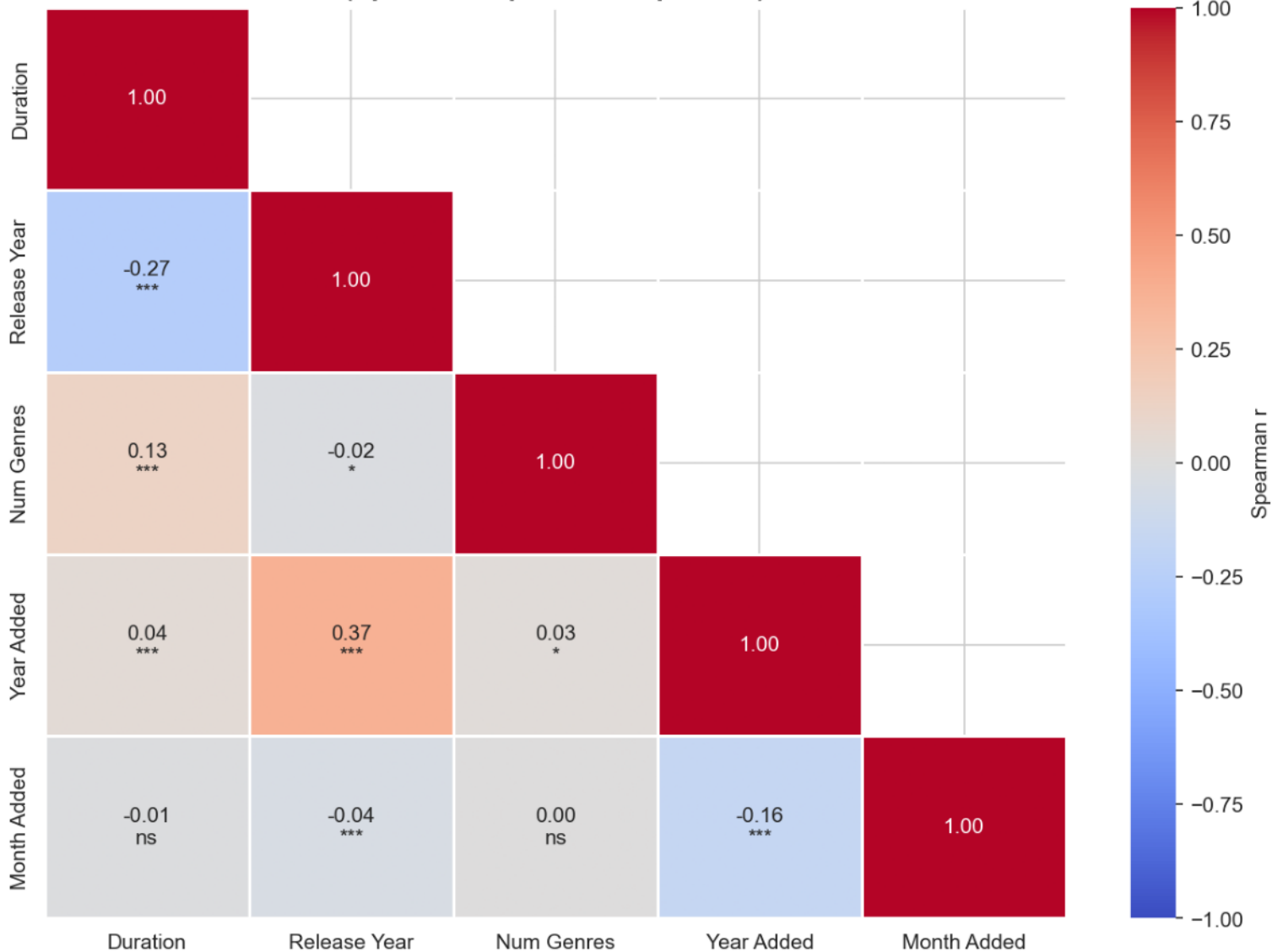
Statistical Summary of Numeric Features — Movies vs TV Shows

Feature	Type	N	Mean	Median	Std Dev	Skewness	Kurtosis	Min	Max
Duration Value	Movies	6112	99.54	98.00	28.30	0.20	2.30	3	312
Duration Value	TV Shows	2663	1.76	1.00	1.58	3.42	16.13	1	17
Release Year	Movies	6112	2013.14	2016.00	9.64	-3.05	12.14	1942	2021
Release Year	TV Shows	2663	2016.59	2018.00	5.75	-5.42	53.48	1925	2021
Num Genres	Movies	6112	2.15	2.00	0.78	-0.27	-1.32	1	3
Num Genres	TV Shows	2663	2.29	2.00	0.78	-0.56	-1.15	1	3

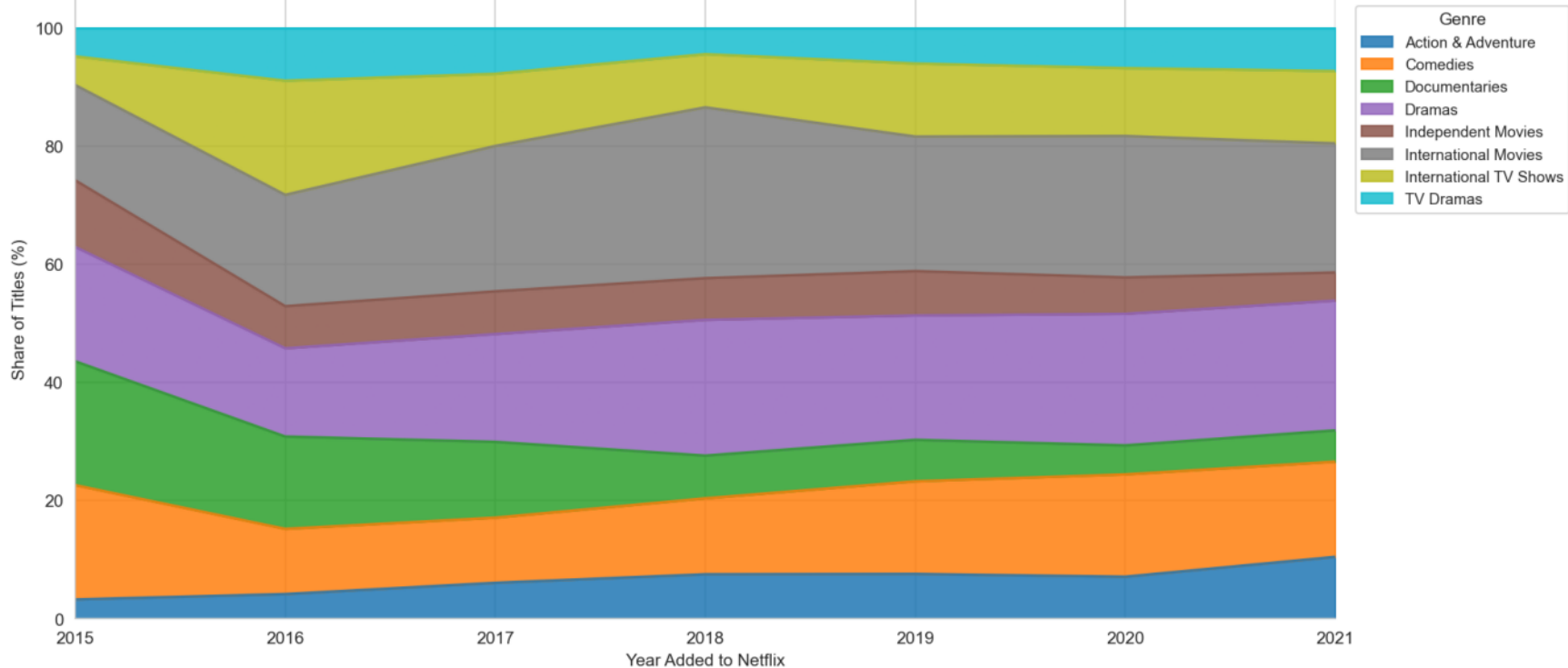
Distribution Analysis of Key Features



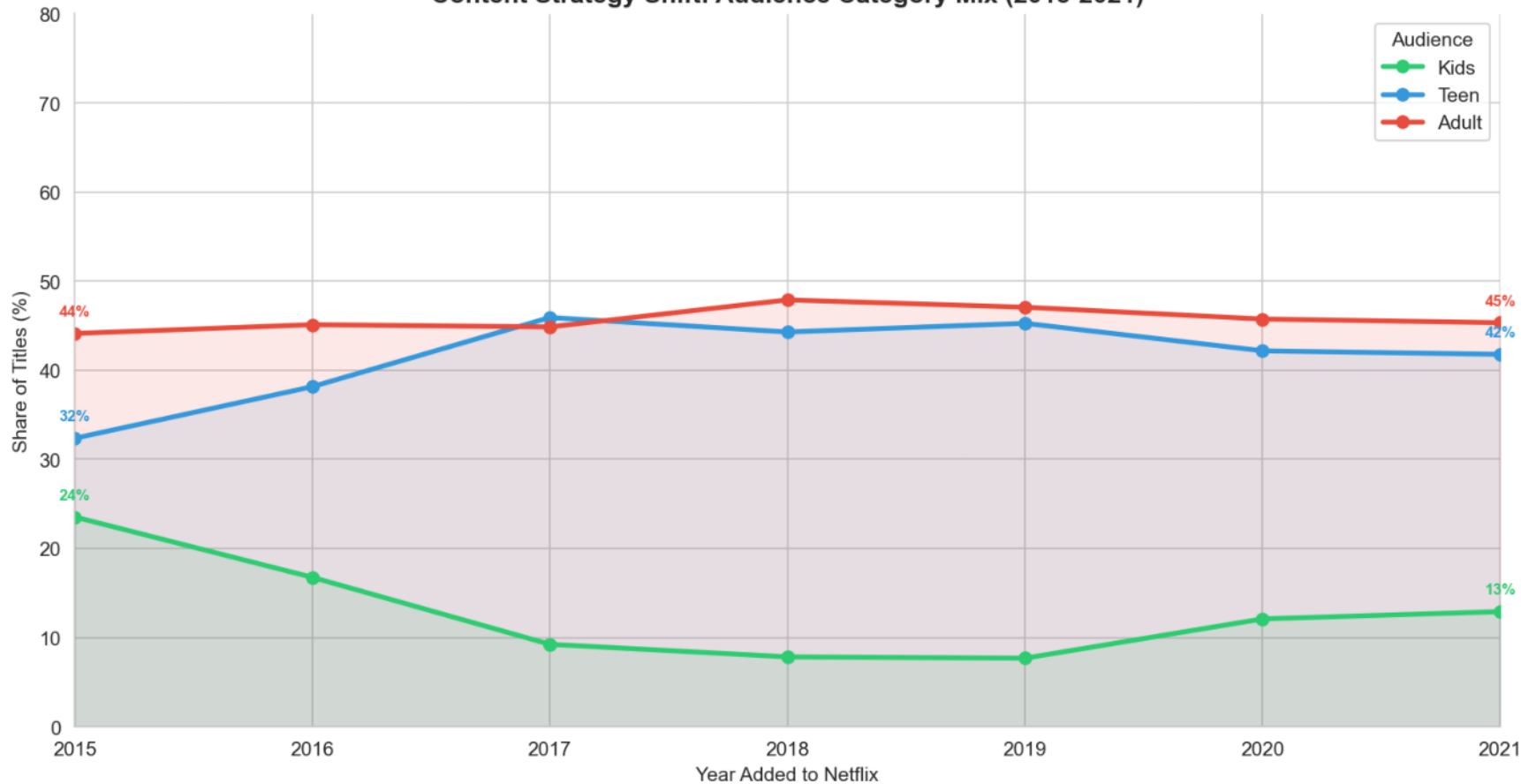
Spearman Correlation Matrix
 (* p<0.05 ** p<0.01 *** p<0.001)



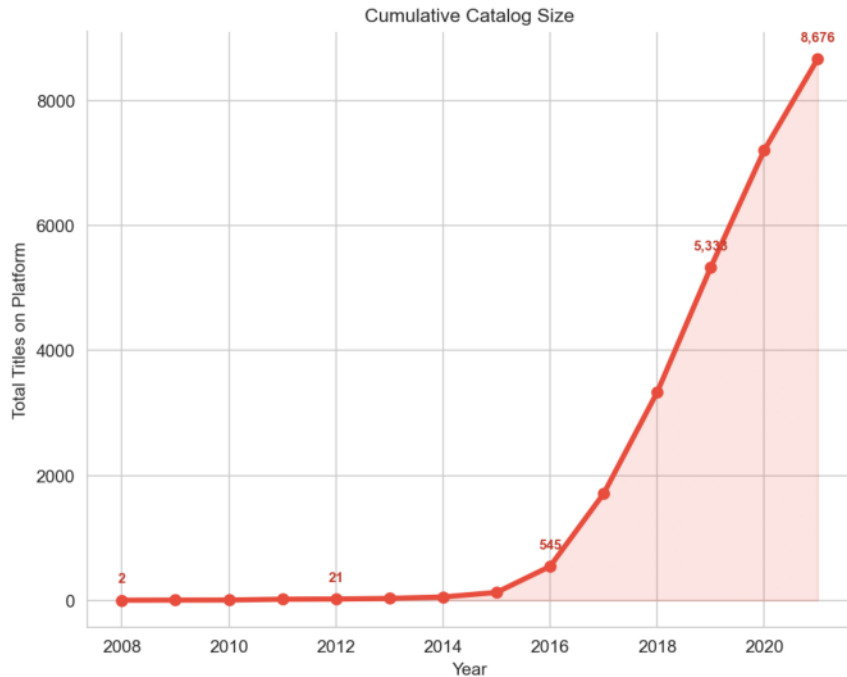
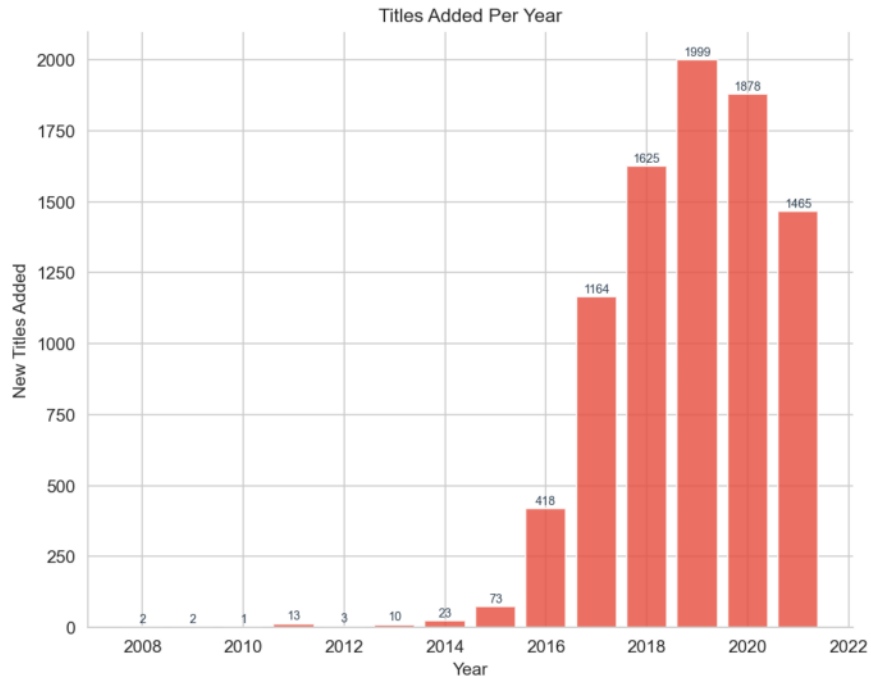
Top Genre Share Evolution (2015-2021, Normalized %)



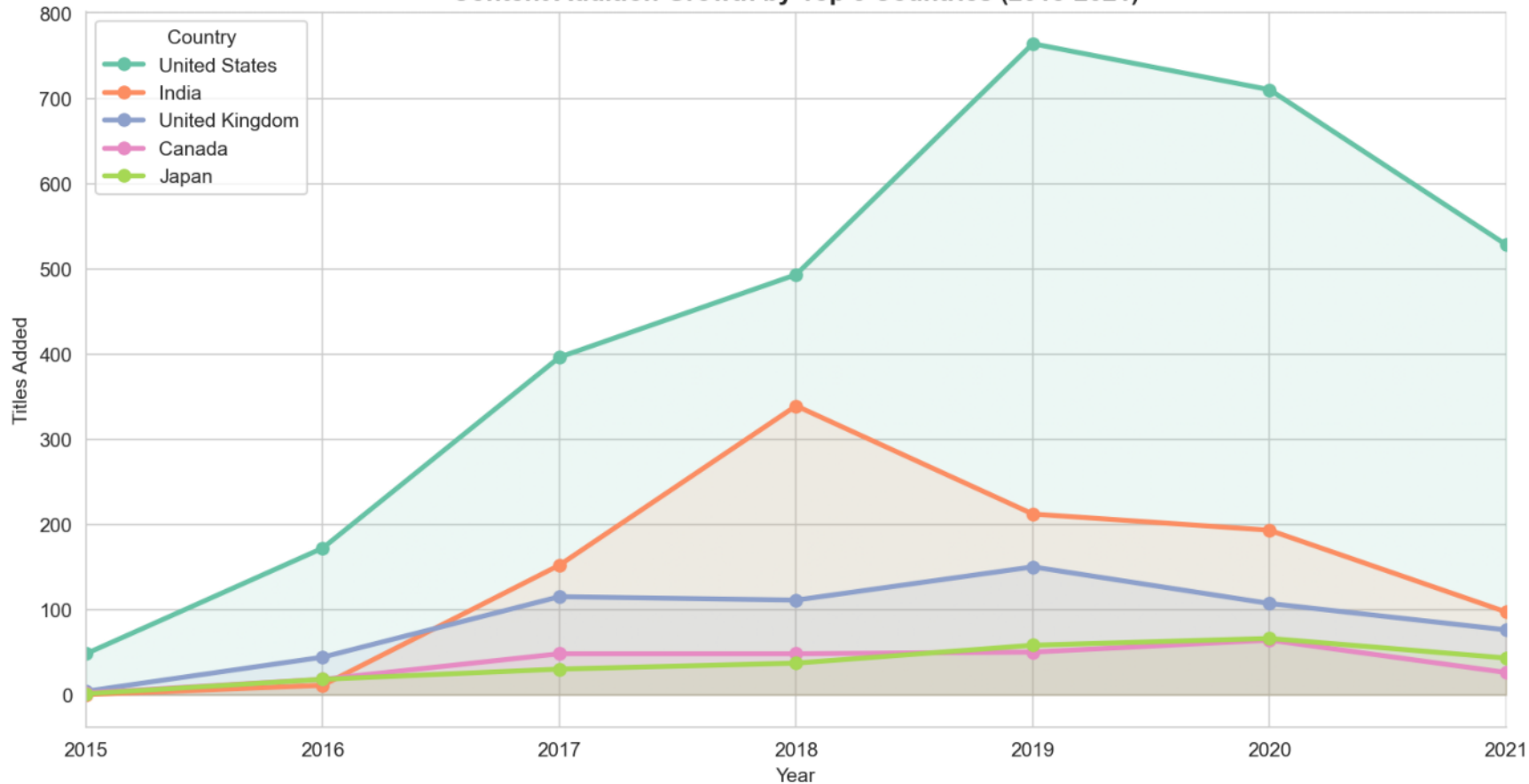
Content Strategy Shift: Audience Category Mix (2015-2021)

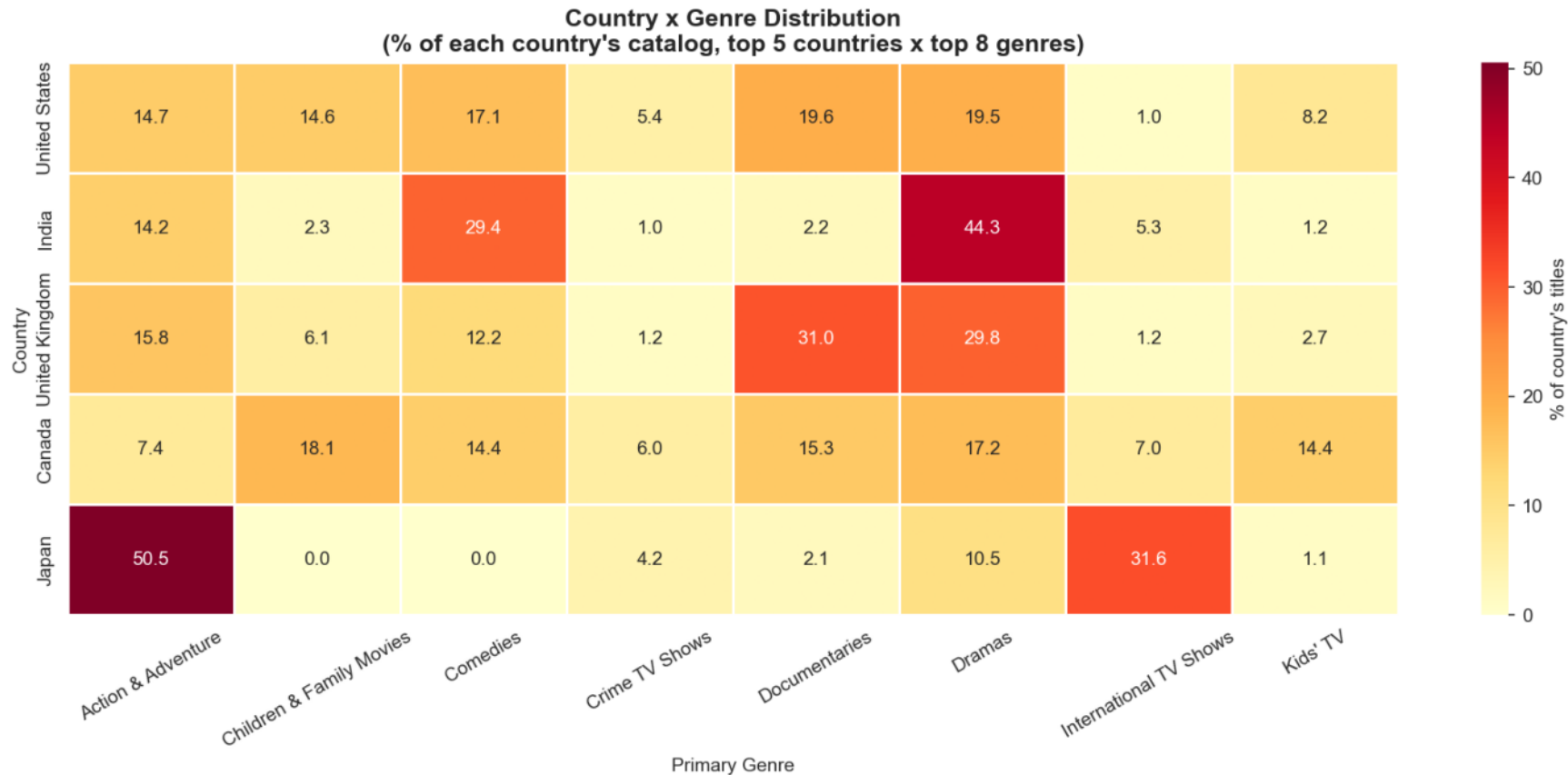


Netflix Catalog Growth Over Time

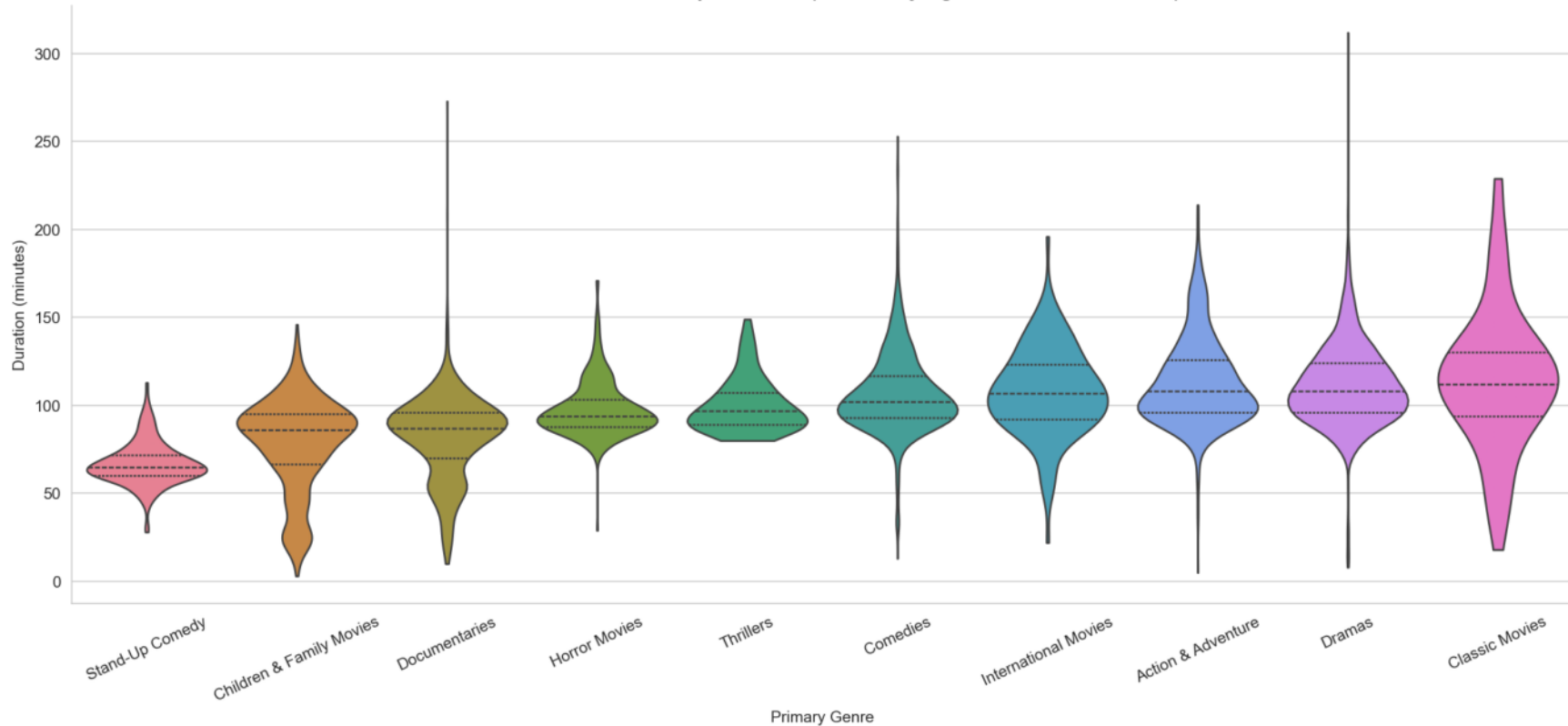


Content Addition Growth by Top 5 Countries (2015-2021)

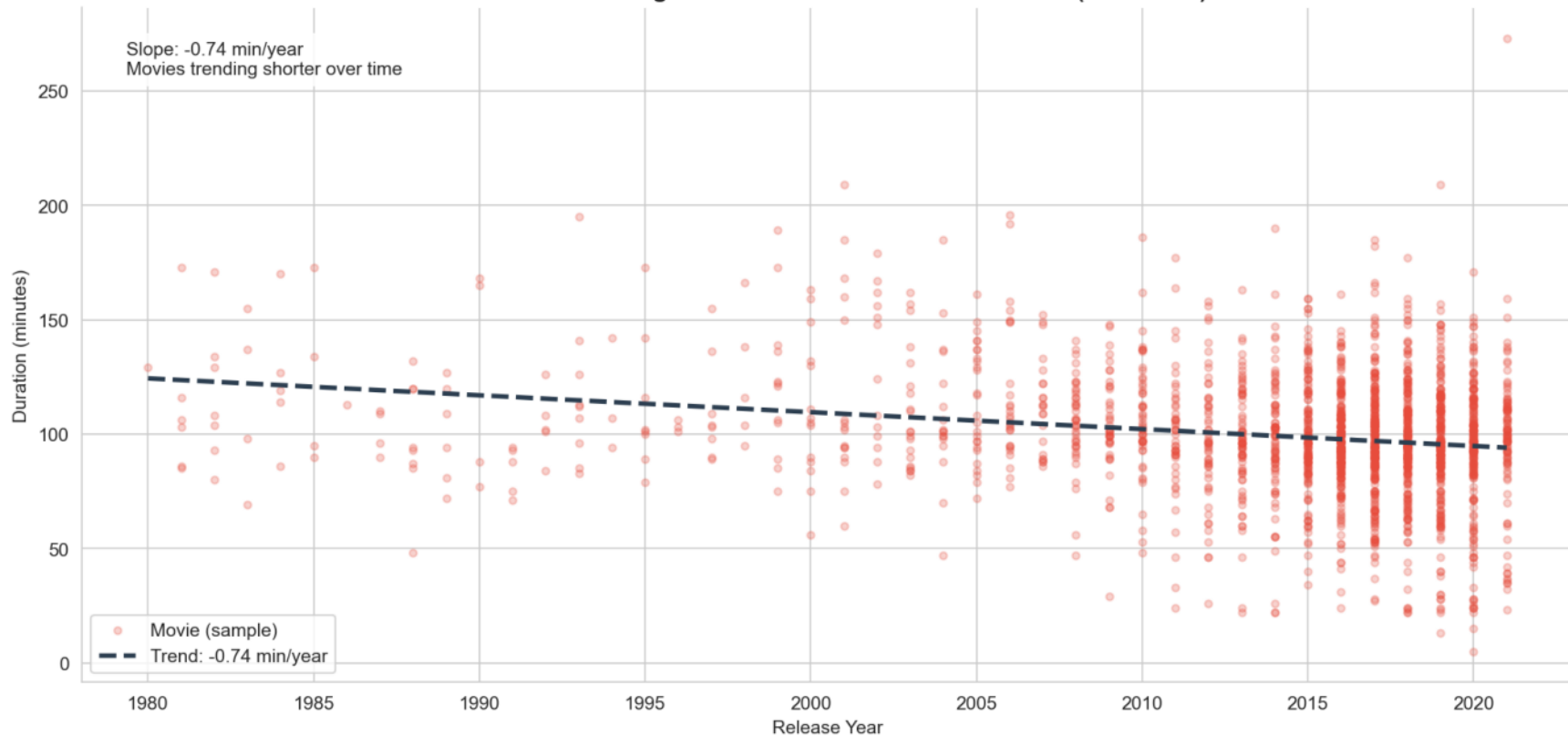




Movie Duration Distribution by Genre
Kruskal-Wallis H=1986.7, $p=0.00e+00$ (statistically significant differences exist)

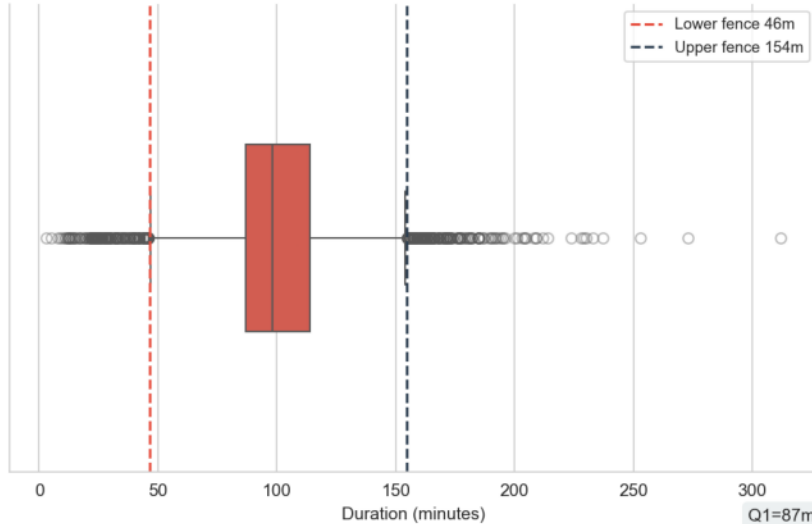


Are Movies Getting Shorter? Release Year vs Duration (1980-2021)



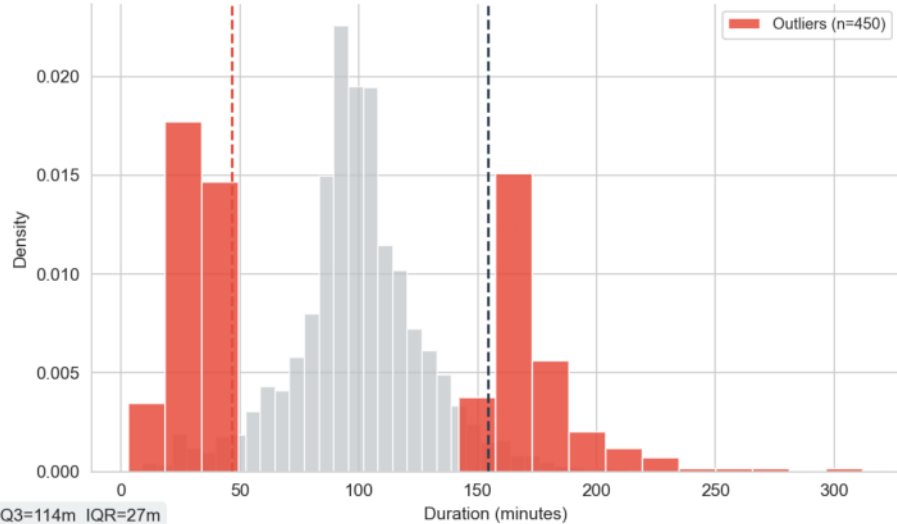
Movie Duration Outlier Analysis

Boxplot — 450 outliers detected (7.4%)

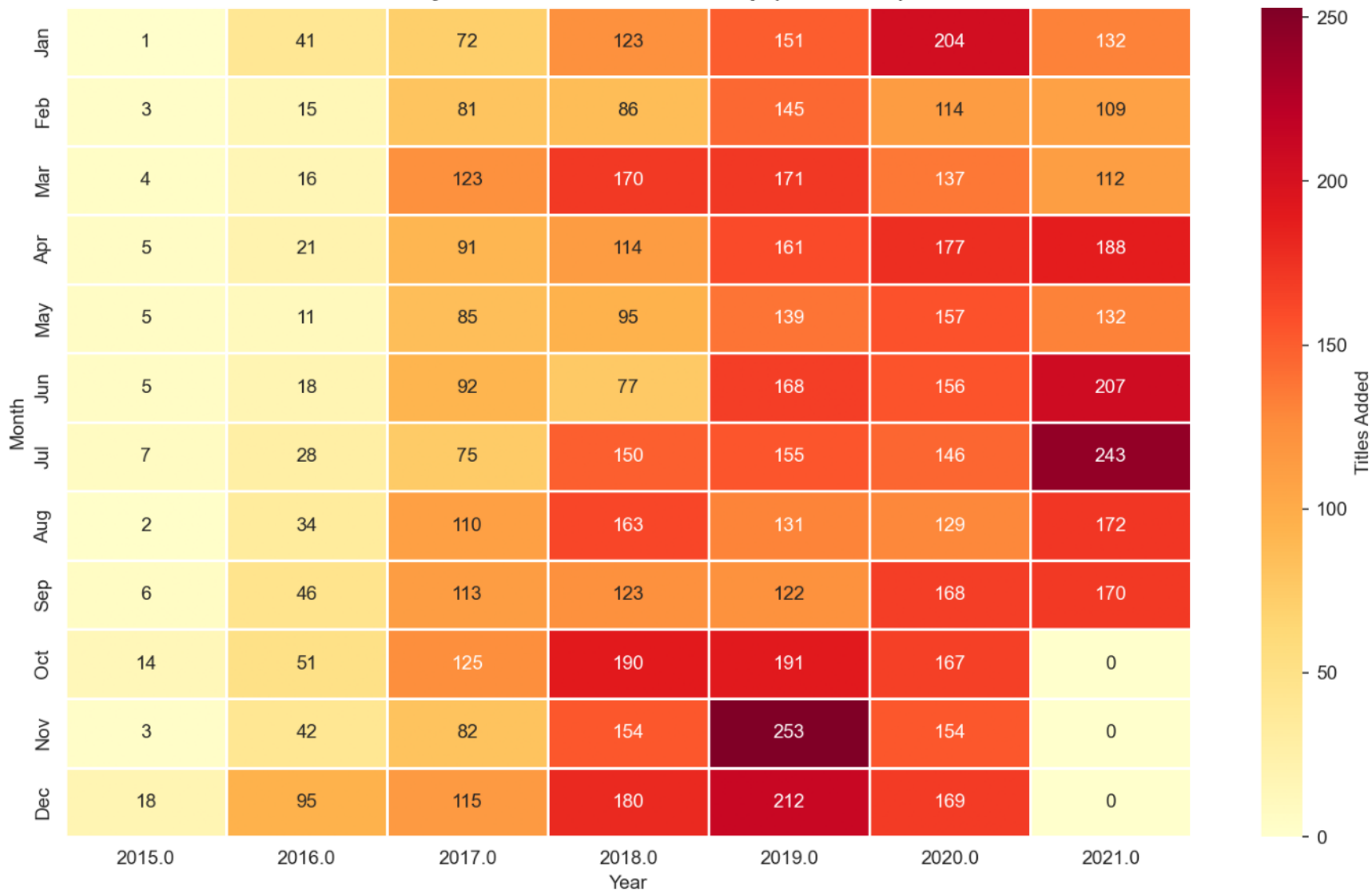


Q1=87m Q3=114m IQR=27m
Normal range: [46, 154] min
Outliers: 450 titles (7.4%)

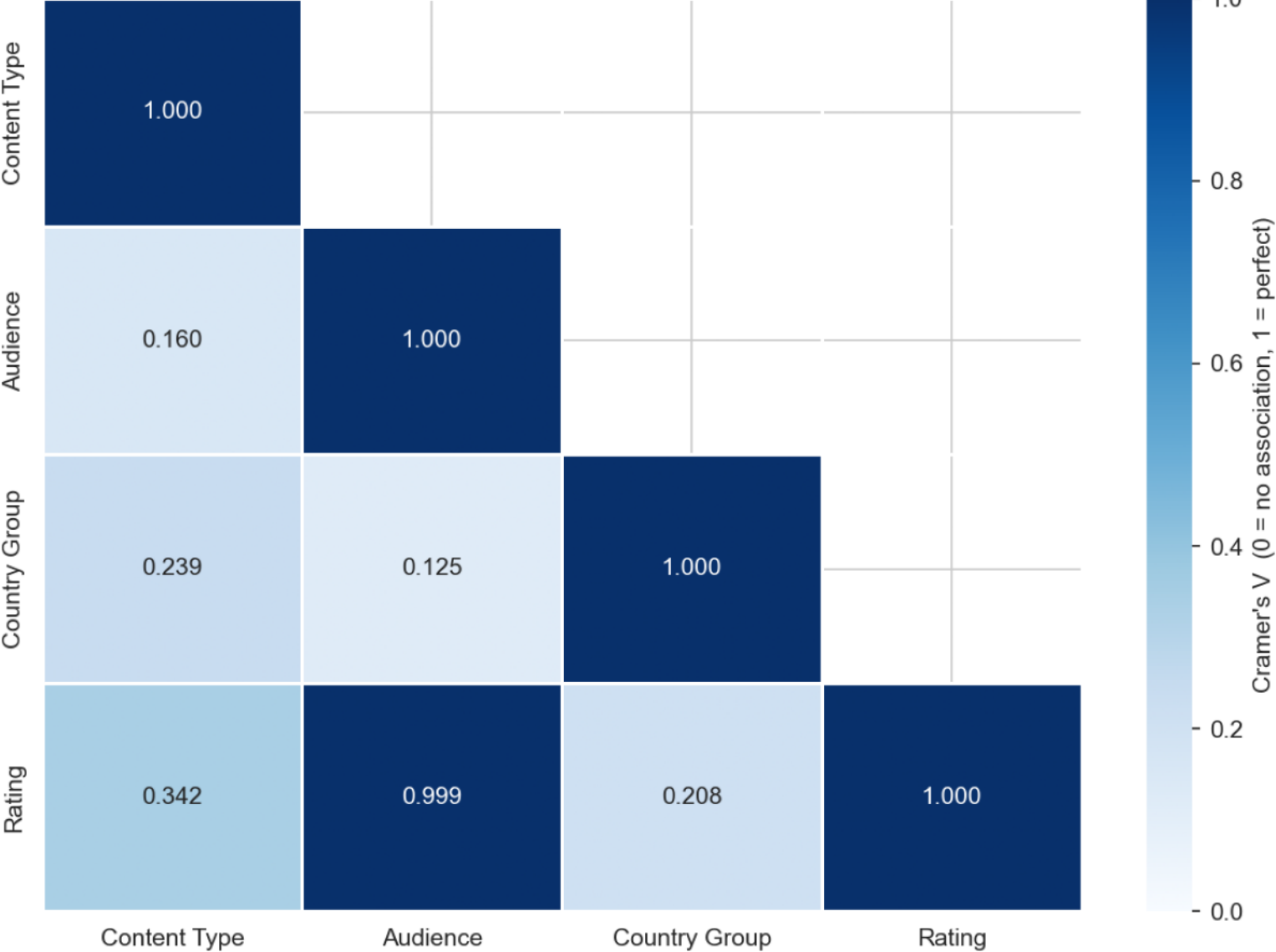
Distribution — Normal range vs Outliers



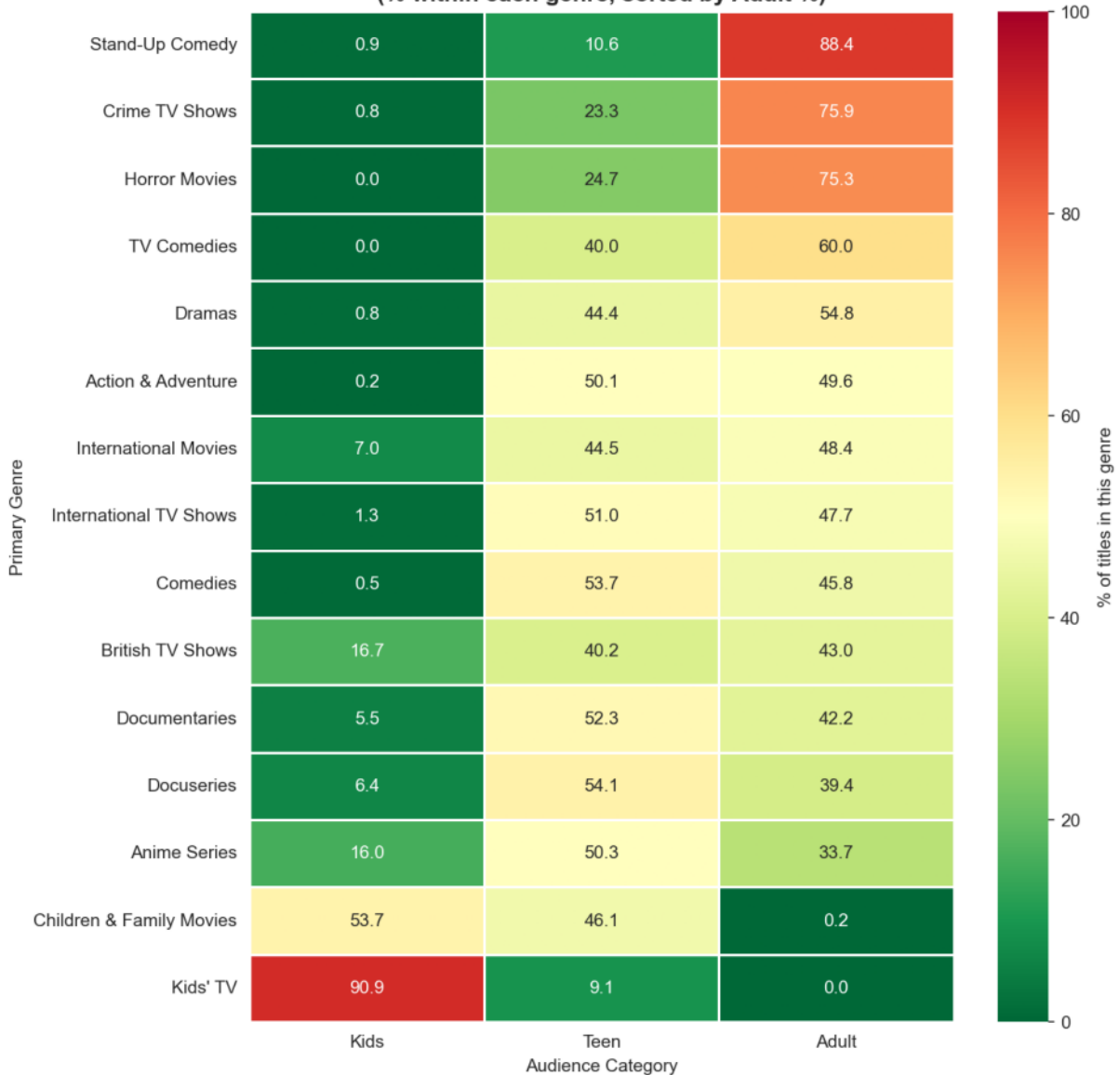
Monthly Content Addition Heatmap (2015-2021)



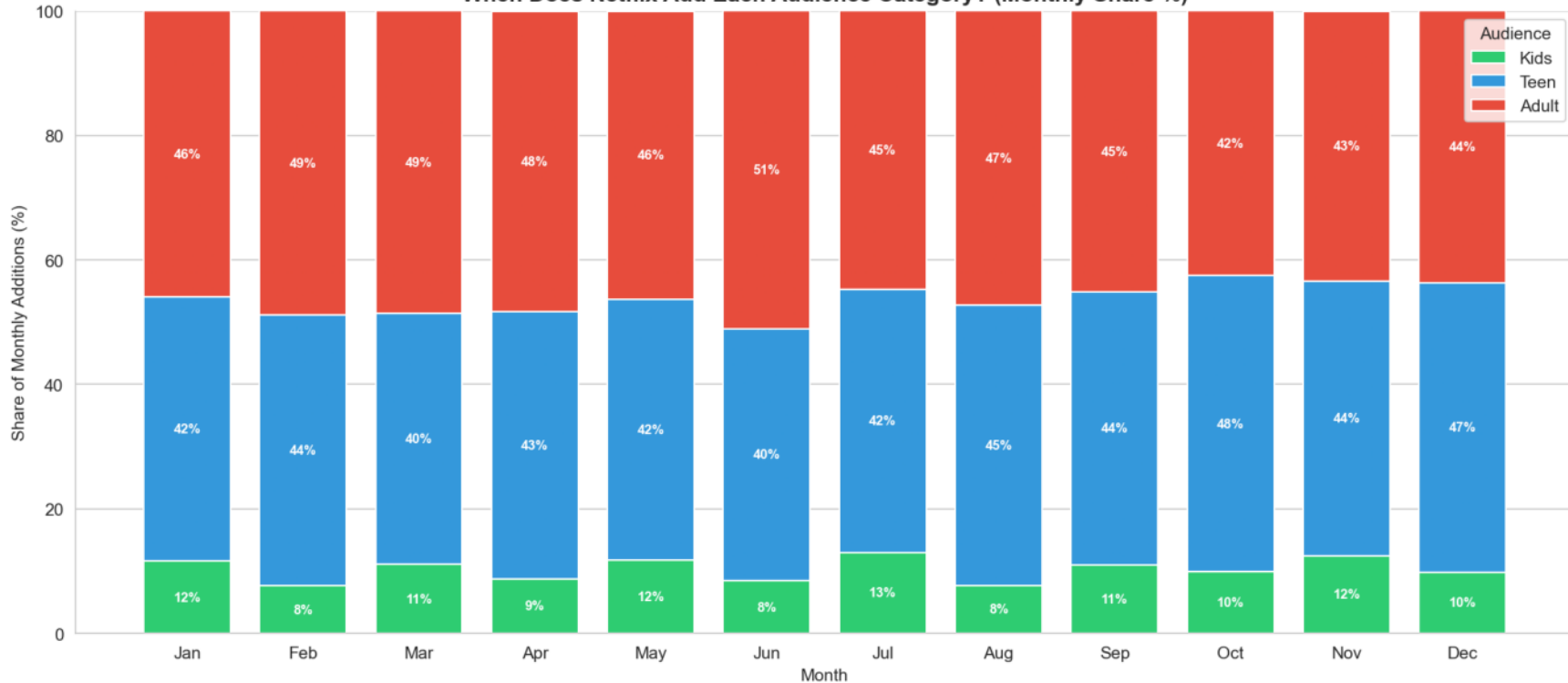
Cramer's V Association Matrix Between Categorical Features



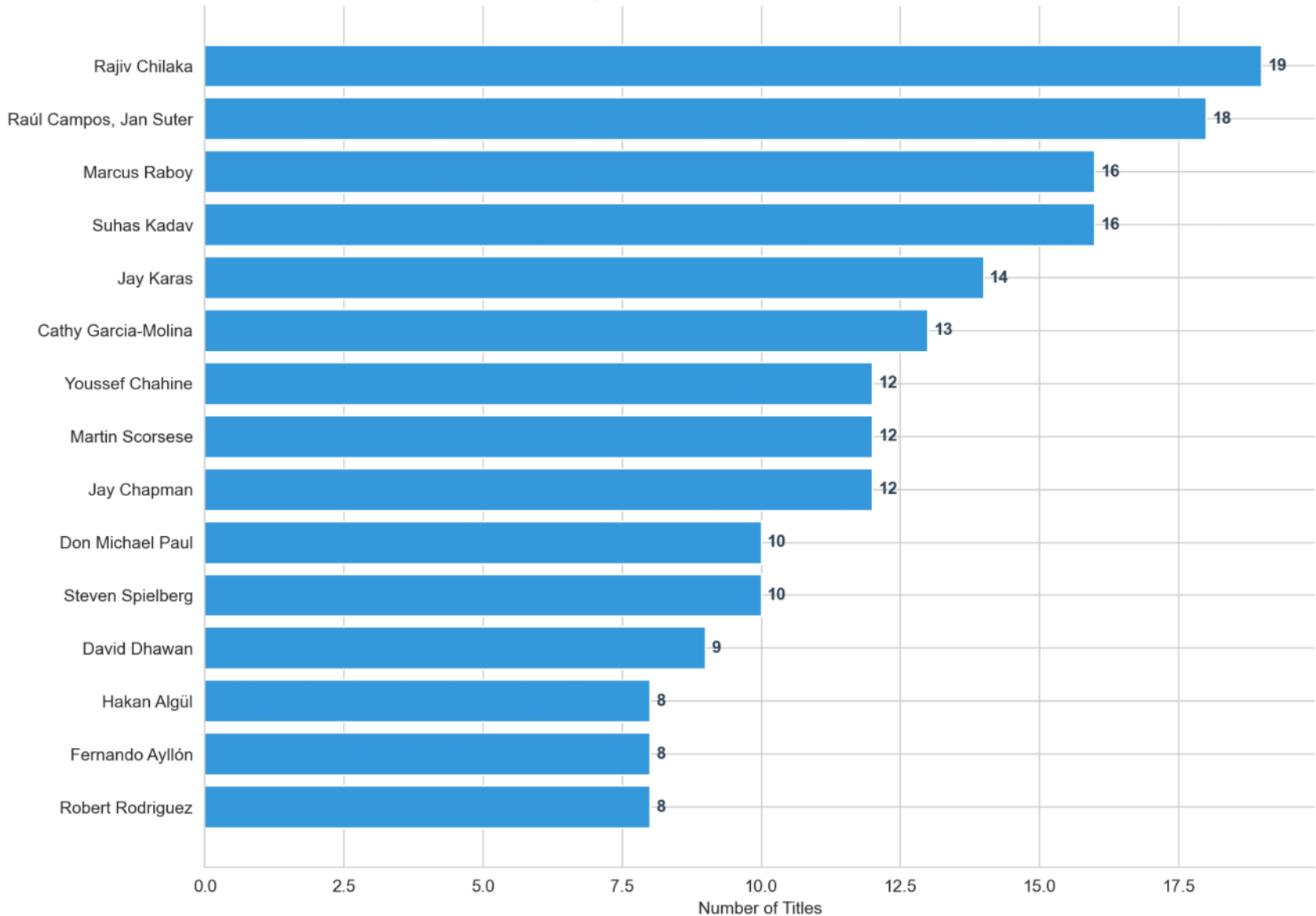
Genre x Audience Distribution
 (% within each genre, sorted by Adult %)

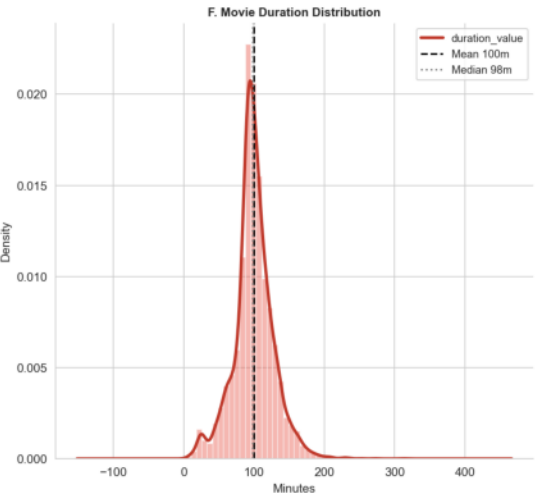
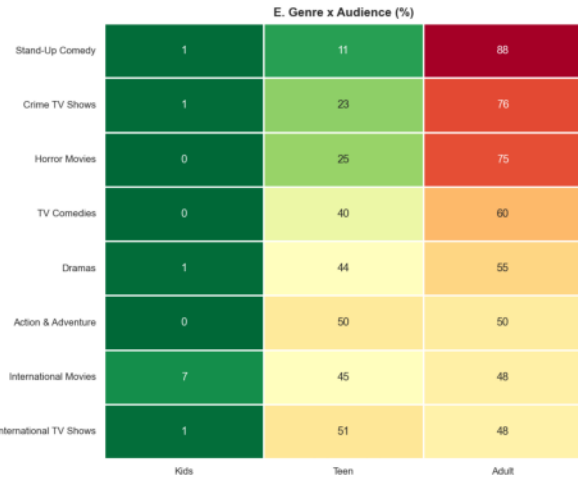
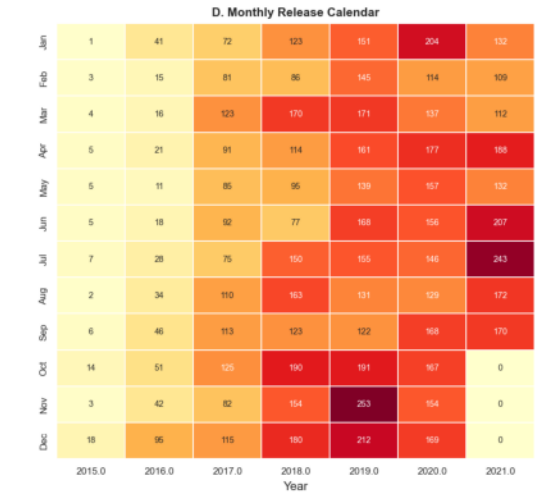
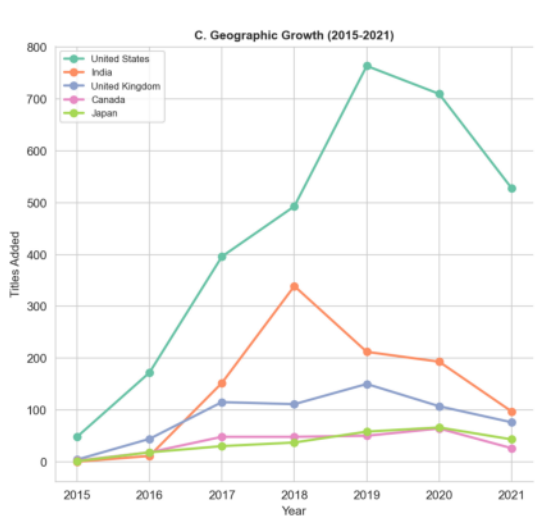
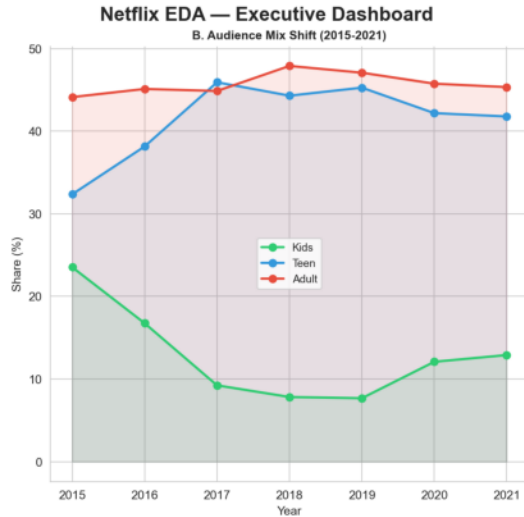
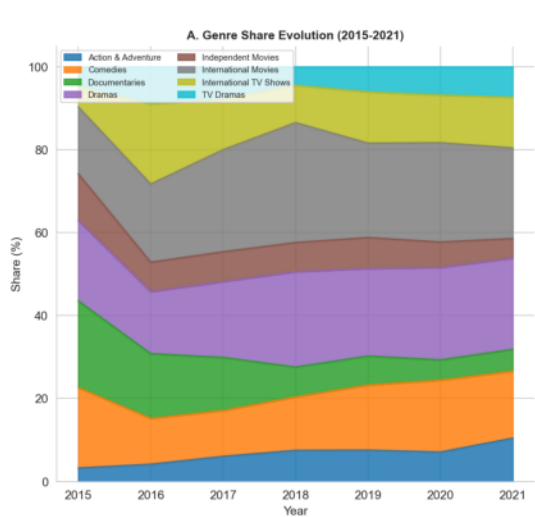


When Does Netflix Add Each Audience Category? (Monthly Share %)



Top 15 Most Prolific Netflix Directors





Netflix EDA — Findings & Recommendations

Key Findings Summary

Fastest growing genre (2015-2021): International TV Shows (+7.4 pp)

Kids content: 23.5% -> 12.9% (halved) – Teen: 32.4% -> 41.8%

India grew fastest among content-producing countries

Peak addition month: Jul

Stand-Up Comedy: 88% Adult – most audience-specific genre

Kruskal-Wallis confirms genre significantly affects movie duration (H=1987)

Duration trend: -0.74 min/year (movies trending shorter)

Outliers: 450 movies outside IQR bounds – genuine edge cases, not errors

Recommendations

1. Double down on International TV Shows – fastest growing genre
2. Monitor Kids content decline – family segment at risk
3. Expand India partnerships – highest growth market
4. Leverage July surge for major title launches
5. Stand-Up Comedy is a high-retention Adult niche